Final Group Project

Due: Fri Oct 2, 2026 11:59pm

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Assignment

This is a group assignment. Your group will identify a concrete problem that you can argue would matter to real business and make a technical contribution toward solving it with machine learning or AI. You will present your work to the class in a ten-minute presentation, and the class's evaluation of your work will count for half of the project grade.

The assignment is intentionally open-ended. You may build a predictive model, a retrieval-augmented generation system, an AI agent, a fine-tuned model, an evaluation of an existing AI system, or something else entirely. You may choose any industry, any dataset, and any company (real or realistic). There are two requirements.

The project must contain a technical contribution. You must build, train, or analyze something, and you must evaluate it. Every project needs some answer to the question "how well does this work?" a held-out test set, a hand-labeled sample, a set of verified test questions, a comparison against human performance, or a backtest, as appropriate to the project. A presentation about what AI could do for a company, without a working artifact behind it, does not satisfy this requirement. Neither does a demonstration that consists of prompting Claude or ChatGPT, with no data, no system built around the model, and no evaluation. Using AI tools to build your project is expected and encouraged; the project itself must be something you built and measured.

The problem must be concrete and business-relevant. As part of the project, you need to describe who would be the user of your output (whatever the output looks like) and why they would derive value from it. It is not enough to have a theoretically interesting problem, you need to have some specific business relevant application in mind and you need to communicate that your output would meaningfully contribute to that application.

A rigorous project that concludes an approach does not work, and shows why, can earn a high grade. I am grading the quality of the investigation and the strength of the evidence, not whether the results came out positive.

Directions to consider

These are general examples of the kinds of directions to consider. Each notes what the technical contribution and evaluation would look like.

- **A predictive model that supports a decision.** Supervised learning on a real dataset — churn, default, demand, cancellations — with careful validation, an honest error analysis, and a decision analysis: what threshold, what costs, what changes when it is deployed. Evaluation: held-out performance plus a cost-weighted decision metric.
- **A RAG system over a real document corpus.** Filings, manuals, contracts, policies — grounded answers with citations. Evaluation: a gold question set that you build and verify by hand, scored on answer accuracy and citation correctness, with an analysis of whether failures come from retrieval or from generation.
- **An agent that automates an analytical workflow.** Tools over a database, an API, or documents, with auditable transcripts. Evaluation: a set of tasks with verified answers, task success measured by type, and an account of the failure modes.
- **A fine-tuned specialist model.** Teach a small open-weight model a narrow, valuable skill and compare it against prompting a frontier model. Evaluation: base model vs. fine-tuned model vs. frontier model on a held-out split, along with what the specialization cost the model elsewhere.
- **An evaluation study of an AI system.** Measure an AI tool against human labels: agreement, chance baselines, and where the human ground truth itself is uncertain. Evaluation: the study is the contribution; the standard is rigor.
- **Something else.** If your idea has a real artifact and a real evaluation feel free to propose it.

Example projects

The sketches below show the expected scope on real, available datasets. You are welcome to use one of the below datasets and problem framings, but I strongly encourage you to explore other options as well.

- **Predicting cancellations to set an overbooking policy.** [Hotel Booking Demand (Kaggle) (https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand)](https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand) — 119,390 real bookings from two Portuguese hotels with 32 features. Build a cancellation model for the hotels' revenue manager, then translate predicted risk into an overbooking policy with explicit costs for a walked guest versus an empty room. Evaluation: held-out model performance plus a policy backtest — revenue under your policy versus "never overbook" and "always overbook a fixed percentage."

- **Demand forecasting where errors have asymmetric costs.** M5 Forecasting (Kaggle) (https://www.kaggle.com/competitions/m5-forecasting-accuracy) — daily sales for 30,490 item-store combinations across ten Walmart stores, with prices and promotions. Forecast at the item-store level for a category manager, with attention to stockout versus holding costs and to which items are effectively unforecastable. Evaluation: a rolling-origin backtest against naive and seasonal baselines, expressed in dollars as well as error metrics.
- **Credit decisioning with relational data.** Home Credit Default Risk (Kaggle) (https://www.kaggle.com/competitions/home-credit-default-risk) — 307,511 applications plus six linked tables of bureau records, prior applications, and payment histories. Build a default model for a lender serving borrowers with thin credit files, and address deployment directly: which features would survive regulatory scrutiny, and how the approval threshold affects different applicant groups. Evaluation: held-out discrimination plus a profit curve and a threshold recommendation.
- **Pricing listings from structured data and text.** Inside Airbnb (https://insideairbnb.com/get-the-data/) — listings, calendars, and reviews for dozens of cities, updated quarterly. Build a nightly-rate recommender for a host-management company in one city, combining tabular features with embeddings of listing text and reviews. Evaluation: held-out pricing error against a tabular-only baseline, plus an analysis of mispriced listings the business could act on.
- **Contract review with expert ground truth.** CUAD (https://www.atticusprojectai.org/cuad) — 510 real commercial contracts with more than 13,000 clause annotations made by lawyers. Build a due-diligence assistant for an M&A team that answers questions like "find the change-of-control clause" with page-cited passages. Evaluation: retrieval hit rate and citation accuracy against the expert annotations, broken out by clause type.
- **An agent over a live public API.** NYC 311 Service Requests (https://data.cityofnewyork.us/Social-Services/311-Service-Requests-from-2010-to-Present/erm2-nwe9) — roughly 40 million service requests, updated daily, with a public query API. Build an operations-analyst agent for a city department with tools for the 311 API and a local reference table, answering questions like "where are noise-complaint response times deteriorating?" Evaluation: a task set with answers verified by hand-written queries, success measured by task type, with auditable transcripts.
- **A fine-tuned specialist for contract clauses.** LEDGAR (Hugging Face, lex_glue) (https://huggingface.co/datasets/coastalcph/lex_glue) — 80,000 contract provisions from SEC filings labeled across 100 clause types. Fine-tune a small open model to classify clauses for a contract-management product, against the honest baseline of a frontier model with few-shot prompting. Evaluation: three-way accuracy on the held-out split, cost per thousand clauses for each option, and what the specialist lost on general tasks.

- **Routing customer support with a model you own.** [Bitext customer support (https://huggingface.co/datasets/bitext/Bitext-customer-support-llm-chatbot-training-dataset)](https://huggingface.co/datasets/bitext/Bitext-customer-support-llm-chatbot-training-dataset) (27,000 utterances, 27 intents) or [Banking77 (https://huggingface.co/datasets/PolyAI/banking77)](https://huggingface.co/datasets/PolyAI/banking77) (13,000 real banking queries, 77 intents). Build the intent router for a support organization and compare owning a fine-tuned model against calling an API. Evaluation: held-out accuracy and the confusion structure — which intents are genuinely confusable, and what that means for queue design.
- **An evaluation study of machine labels against human labels.** [GoEmotions (https://huggingface.co/datasets/google-research-datasets/go_emotions)](https://huggingface.co/datasets/google-research-datasets/go_emotions) — 211,000 individual rater annotations, with rater IDs, across 58,000 comments — or the [CFPB consumer complaint database (https://www.consumerfinance.gov/data-research/consumer-complaints/)](https://www.consumerfinance.gov/data-research/consumer-complaints/) — millions of complaints with human-assigned product labels. Measure how well an LLM labeler agrees with humans, and how well humans agree with each other, which sets the ceiling. Evaluation: agreement statistics with chance baselines, and accuracy broken out by the level of human agreement.

Proposal

Due Tuesday, September 8, one page, submitted on Canvas (there will be a separate Canvas assignment to submit). It should contain: the problem and the business application; the dataset and its source; what you will build or analyze; what your evaluation will be — what you will measure, and against what ground truth; and anything you need from me. The proposal is not graded, but every project must be approved before it is presented. If you would like feedback on an idea before the deadline, come to office hours or email me.

Presentation

Ten minutes per group, in class, during Sessions 13 and 14, with brief Q&A as time allows. Presentation order will be drawn at random and announced after proposals are in. A live demonstration is welcome where it serves the presentation; have a recorded backup in case of technical problems. All group members should be involved in the presentation.

Your audience is the class, and the class will grade you on the questions listed below. Aim the ten minutes at them. You should not spend most of your time on technical details, except when they are illuminating for understanding why your approach does or does not work (neither I nor your class really care what the exact tuning parameters were for your CatBoost model, for example, but I do care if you found text embeddings to be an important component of your data pipeline). You will also submit artifacts that document your approach (Juptyer notebooks, python files, a writeup), and the technical details should be there.

Grading

Your group's project score is an equal blend of my evaluation and the class's evaluation:

Project Score = 0.5 × Instructor Score + 0.5 × Class Score

Instructor score

- **Technical contribution, 30%:** You built, trained, or analyzed something real, with sound methods, sensible choices, and honest treatment of limitations. As in past years, my expectations scale with the difficulty of what you attempted.
- **Evaluation, 25%:** You designed an appropriate test and ran it: suitable ground truth, honest baselines, correct interpretation. An error analysis — where the system fails and why — is expected in strong projects.
- **Communication, 20%:** The presentation is clear and well-constructed, and the notebook is annotated well enough that I can follow every decision without asking you. Reading other people's code is difficult, so err on the side of more explanation.
- **Business framing, 15%:** A named stakeholder with a real decision, actionable recommendations, sound economics, and no hidden assumptions that would prevent the work from being applied in practice.
- **Ambition and novelty, 10%:** The project goes somewhere the starter code does not already go — an unusual problem, a creative evaluation, a surprising or well-defended negative finding.

Class score

After each presentation, every student outside the presenting group rates it on three questions, each on a 1–5 scale:

1. **Problem:** "This team convinced me their problem is real and worth solving."
2. **Insight:** "I took away a genuine insight about what works, what does not, or why."
3. **Evidence:** "The team supported its conclusions with evidence."

These questions are published in advance so that you can build your presentation around them.

The mechanics: ratings are submitted through a short form during the presentation sessions and close at midnight the same day. Submitting ratings for every group you watched is required and counts toward class participation. Your three ratings for a group are averaged into a single score. Because raters differ in how generously they score, each rater's scores are standardized (using that rater's own mean and standard deviation across all groups they rated) before averaging — the same z-score idea we use elsewhere in the course. The standardized ratings are then averaged across raters for each group and mapped onto the same scale as the instructor score. Averaging over the whole class reduces the influence of any single rating, and I will discard ballots that are clearly not made in good faith.

Individual score

After submission, each of you will privately allocate 100% of the project effort across your teammates, excluding yourself. For a group of n people, your final score is:

Final Individual Score = 0.5 × Project Score + 0.5 × (n – 1) × (your average rated effort share) × Project Score

Everyone receives at least half of the project score. If your teammates rate you at an equal share — 1/(n–1) — your score equals the project score; above an equal share, it is higher; below, lower. This is the same formula as in prior years, written so that it applies to any group size.

Deliverables

One zip file named "Group_#.zip," uploaded to a Box link on Canvas by October 2 at midnight, containing:

1. The slides used in your presentation.
2. A Jupyter notebook (or small repository) with your full pipeline, annotated in the style of the course starter code so that your reasoning is easy to follow: what you tried, what you kept, and why. If you did not use a Jupyter notebook, you need to include a detailed README.md explaining the structure of the code, what you did, and why. **You should err on the side of over-explaining.** If you think something is clear, someone else reading your code will still find it confusing.
3. Your evaluation: the test data or task set, the ground truth and where it came from, and the code that produces your reported numbers. I should be able to re-run your evaluation.
4. The dataset, or clear access instructions.
5. A one-page summary — problem, approach, findings, recommendation — written for an executive who missed the presentation.
6. A short AI-use note in the notebook: which AI tools you used, for what, and how you verified their output.

If you need to re-upload, mark the newer file clearly and email me to say which submission counts.

Use of AI

Consistent with the course policy: I expect you to build with AI coding agents, and the mechanics of your pipeline may be heavily AI-assisted. You may use the course provided tools (JupyterHub and OpenCode), but you are not restricted to those tools. You can use any AI tool for assisting in your project, but you have to document how you built it in such a way that others could reproduce your work. It is not enough that "Claude told me the answer was x". You need to have clear code that code be run on anyone's computer that shows every step of the process. The judgment must be your own. The choice of problem, the design of the evaluation, the interpretation of the results, and the recommendations are the parts of the project that I and your classmates are grading, and you will answer questions about your project live.

The goal of this project is to carry out the full arc we practice all quarter — problem, data, system, evaluation, judgment, communication — on a problem your group chose because it matters. Pick something you actually want to know the answer to. Good luck, and reach out as you go.